

Advanced Bayesian Learning

Variational inference

Lecture 5

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Topic overview

- Variational inference (VI)
- Mean-field VI
- Stochastic VI
- Fixed form VI
- Stochastic gradients and variance reduction
- Automatic differentiation

Variational inference

■ Literature:

- ▶ *Variational Inference: A Review for Statisticians*, JASA article by Blei et al (2017).
- ▶ *A practical tutorial on Variational Bayes* - notes by Minh-Ngoc Tran at Sydney University.

■ Aim: approximate $p(\boldsymbol{\theta}|\mathbf{y})$ with a (simpler) distribution $q(\boldsymbol{\theta})$.

■ Laplace approximation from optimization:

$$q(\boldsymbol{\theta}) = N \left[\tilde{\boldsymbol{\theta}}, \left(-\nabla \nabla^T \log p(\boldsymbol{\theta}|\mathbf{y})|_{\tilde{\boldsymbol{\theta}}} \right)^{-1} \right]$$

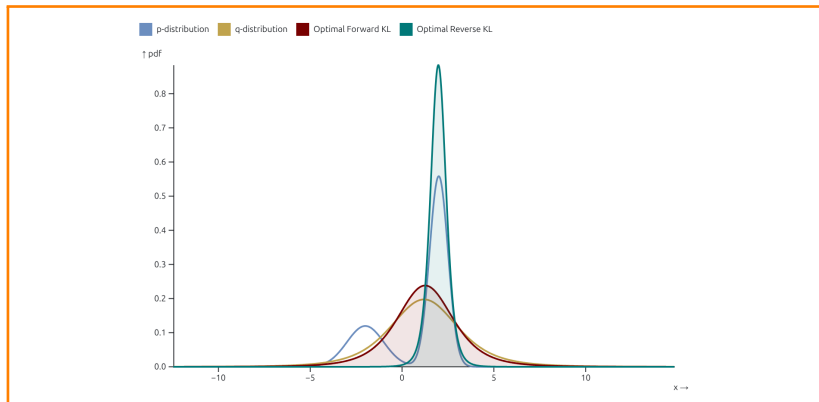
■ Kullback-Leibler divergence of $g(x)$ from $f(x)$

$$\text{KL}(f \| g) = \int \ln \frac{f(x)}{g(x)} f(x) dx = \mathbb{E}_f \left(\ln \frac{f(x)}{g(x)} \right)$$

■ Properties of KL:

- ▶ $\text{KL}(f \| g) \geq 0$
- ▶ $\text{KL}(f \| g) \neq \text{KL}(g \| f)$ in general. First density is the judge.

Interactive - KL divergence



Variational inference

- **VI**: approximate $p(\boldsymbol{\theta}|\mathbf{y})$ by $q(\boldsymbol{\theta}) \in \mathcal{Q}$

$$q^*(\boldsymbol{\theta}) = \arg \min_{q(\boldsymbol{\theta}) \in \mathcal{Q}} \text{KL}(q \| p) = \int q(\boldsymbol{\theta}) \ln \frac{q(\boldsymbol{\theta})}{p(\boldsymbol{\theta}|\mathbf{y})} d\boldsymbol{\theta}$$

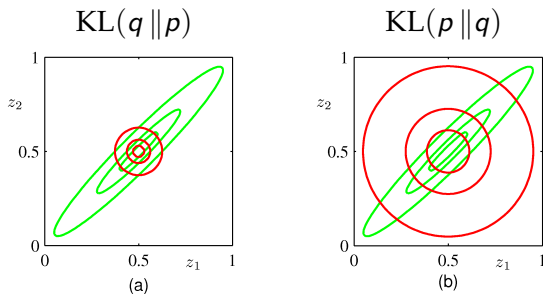
- Turns an inference problem, $p(\boldsymbol{\theta}|\mathbf{y})$, into **optimization**.

- **Ideal**:

- ▶ let $\mathcal{Q}$ be **large enough to approx** $p(\boldsymbol{\theta}|\mathbf{y})$ well
- ▶ let $\mathcal{Q}$ be **small enough for efficient optimization**

- **Early VI**: use restrictive $\mathcal{Q}$ and live with poor approximation.
 - Location of $q(\boldsymbol{\theta})$ is fairly correct.
 - Underestimates the variance (sometimes badly).
- **Modern VI**: use larger $\mathcal{Q}$ + better optimization algorithms + stochastic gradients.

KL - forward or reverse¹



Green contours = True Gaussian posterior
Red contours = Circular Gaussian approximation

¹From Bishop's book *Pattern Recognition and Machine Learning*, Springer.

ELBO - evidence lower bound

- $\text{KL}(q \parallel p)$ is intractable when $p(\theta|\mathbf{y})$ is intractable, but

$$\begin{aligned}\text{KL}(q \parallel p) &= \int q(\theta) \ln \frac{q(\theta)}{p(\theta|\mathbf{y})} d\theta = \int q(\theta) \ln \frac{p(\mathbf{y})q(\theta)}{p(\mathbf{y}|\theta)p(\theta)} d\theta \\ &= \int \ln p(\mathbf{y}) q(\theta) d\theta - \int q(\theta) \ln \frac{p(\mathbf{y}|\theta)p(\theta)}{q(\theta)} d\theta\end{aligned}$$

- Hence $\text{KL}(q \parallel p) = \ln p(\mathbf{y}) - \text{ELBO}(q)$

- $\text{ELBO}(q)$ **evidence lower bound**

$$\text{ELBO}(q) \stackrel{\text{def}}{=} \int q(\theta) \ln \frac{p(\mathbf{y}|\theta)p(\theta)}{q(\theta)} d\theta$$

- Minimizing $\text{KL}(q \parallel p)$ wrt $q \iff$ maximizing $\text{ELBO}(q)$ wrt q .

- ELBO is **lower bounds the (log) marginal likelihood** $p(\mathbf{y})$

$$\text{KL}(q \parallel p) \geq 0 \implies \text{ELBO}(q) \leq \ln p(\mathbf{y})$$

Mean field approximation

- **Mean field VI** is based on factorized approximation:

$$q(\boldsymbol{\theta}) = \prod_{j=1}^p q_j(\theta_j)$$

- **No specific functional forms** are assumed for the $q_j(\theta_j)$.
- **Optimal densities** can be shown to satisfy (MNT Notes):

$$q_j(\theta) \propto \exp \left(E_{-\theta_j} \ln p(\mathbf{y}, \boldsymbol{\theta}) \right)$$

where $E_{-\theta_j}(\cdot)$ is the expectation with respect to $\prod_{k \neq j} q_k(\theta_k)$.

- **Structured mean field approximation**. Group subset of parameters in tractable blocks. Similar to Gibbs sampling.

Mean field VI - algorithm

■ Initialize: $q_2^*(\theta_2), \dots, q_M^*(\theta_p)$

■ Repeat until convergence:

▶ $q_1^*(\theta_1) \leftarrow \frac{\exp[E_{-\theta_1} \ln p(y, \theta)]}{\int \exp[E_{-\theta_1} \ln p(y, \theta)] d\theta_1}$

▶ $q_2^*(\theta_2) \leftarrow \frac{\exp[E_{-\theta_2} \ln p(y, \theta)]}{\int \exp[E_{-\theta_2} \ln p(y, \theta)] d\theta_2}$

▶ $\vdots$

▶ $q_p^*(\theta_p) \leftarrow \frac{\exp[E_{-\theta_p} \ln p(y, \theta)]}{\int \exp[E_{-\theta_p} \ln p(y, \theta)] d\theta_p}$

■ **No assumptions about parametric form** of the $q_j(\theta_j)$.

■ Optimal $q_j(\theta_j)$ often **turn out** to be known distributions.

■ **Just update hyperparameters** in the optimal densities.

Mean field VI

- Alternative formulation that connects to **Gibbs sampling**

$$q_j^*(\theta_j) \propto \exp [E_{-\theta_j} \ln p(\theta_j | \boldsymbol{\theta}_{-j}, \mathbf{y})]$$

where $p(\theta_j | \boldsymbol{\theta}_{-j}, \mathbf{y})$ is the full conditional posterior of θ_j .

- **Structured mean field VI**. Group parameters in tractable blocks.
- Make life easy. When deriving $q_{\theta_1}^*(\theta_1)$:
 - ▶ ignore additive terms in $\ln p(\theta_1, \theta_2, \theta_3, \mathbf{y})$ not involving θ_1 .
 - ▶ mean-field: $\mathbb{E}_{-\theta_1} f(\theta_2)g(\theta_3) = \mathbb{E}_{q_2(\theta_2)} f(\theta_2) \cdot \mathbb{E}_{q_3(\theta_3)} g(\theta_3)$.
 - ▶ And of course $\mathbb{E}_{-\theta_1} f(\theta_1) = f(\theta_1)$

Mean field approximation - Normal model

- **Model:** $X_i | \theta, \sigma^2 \stackrel{iid}{\sim} N(\theta, \sigma^2)$.
- **Prior:** $\theta \sim N(\mu_0, \tau_0^2)$ independent of $\sigma^2 \sim \text{Inv} - \chi^2(\nu_0, \sigma_0^2)$.
- **Mean-field approximation:** $q(\theta, \sigma^2) = q_\theta(\theta) \cdot q_{\sigma^2}(\sigma^2)$.
- Optimal densities

$$q_\theta^*(\theta) \propto \exp \left[E_{q(\sigma^2)} \ln p(\theta, \sigma^2, \mathbf{x}) \right]$$
$$q_{\sigma^2}^*(\sigma^2) \propto \exp \left[E_{q(\theta)} \ln p(\theta, \sigma^2, \mathbf{x}) \right]$$

Normal model - VB algorithm

■ Variational density for σ^2

$$\sigma^2 \sim \text{Inv} - \chi^2 (\tilde{\nu}_n, \tilde{\sigma}_n^2)$$

$$\text{where } \tilde{\nu}_n = \nu_0 + n \text{ and } \tilde{\sigma}_n^2 = \frac{\nu_0 \sigma_0^2 + \sum_{i=1}^n (x_i - \tilde{\mu}_n)^2 + n \cdot \tilde{\tau}_n^2}{\nu_0 + n}$$

■ Variational density for θ

$$\theta \sim N(\tilde{\mu}_n, \tilde{\tau}_n^2)$$

where

$$\tilde{\tau}_n^2 = \frac{1}{\frac{n}{\tilde{\sigma}_n^2} + \frac{1}{\tau_0^2}}$$

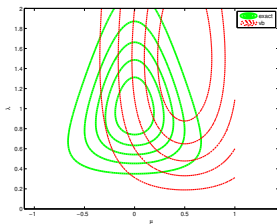
$$\tilde{\mu}_n = \tilde{w} \bar{x} + (1 - \tilde{w}) \mu_0,$$

where

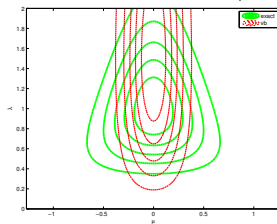
$$\tilde{w} = \frac{\frac{n}{\tilde{\sigma}_n^2}}{\frac{n}{\tilde{\sigma}_n^2} + \frac{1}{\tau_0^2}}$$

Normal example ($\lambda = 1/\sigma^2$)²

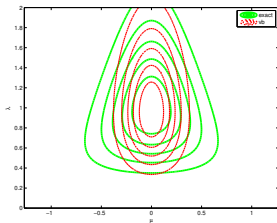
Initial values



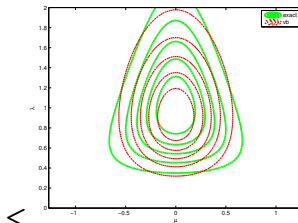
After updating q_μ



After updating q_{σ^2}



At convergence



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²From Bishop's book *Pattern Recognition and Machine Learning*, Springer.

Probit regression³

- **Model:**

$$\Pr(y_i = 1 | x_i) = \Phi(x_i^T \beta)$$

- **Prior:** $\beta \sim N(0, \Sigma_\beta)$. For example: $\Sigma_\beta = \tau^2 I$.

- **Latent variable formulation** with $u = (u_1, \dots, u_n)'$

$$u | \beta \sim N(X\beta, 1)$$

and

$$y_i = \begin{cases} 0 & \text{if } u_i \leq 0 \\ 1 & \text{if } u_i > 0 \end{cases}$$

- Factorized **variational approximation**

$$q(u, \beta) = q_u(u) q_\beta(\beta)$$

³From Ormerod and Wand (2010). *Explaining Variational Approximation*, Amer Stat.

VI for probit regression

■ VI posterior

$$\beta \sim N\left(\tilde{\mu}_\beta, \left(X^T X + \Sigma_\beta^{-1}\right)^{-1}\right)$$

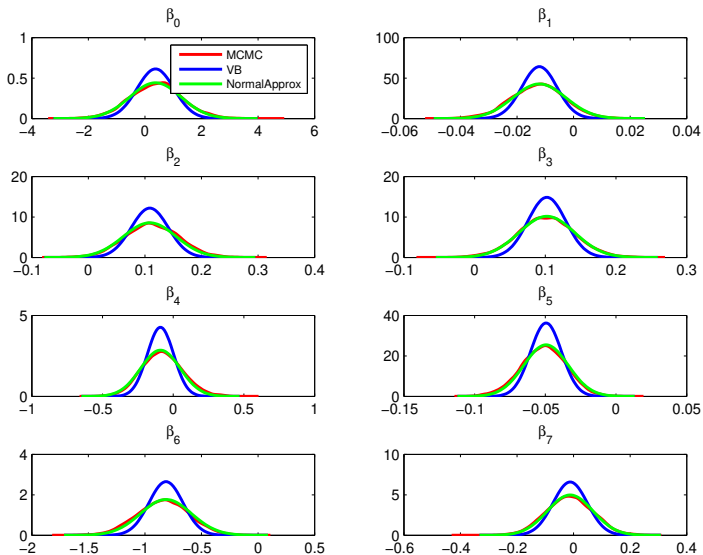
where

$$\tilde{\mu}_\beta = \left(X^T X + \Sigma_\beta^{-1}\right)^{-1} X^T \tilde{\mu}_u$$

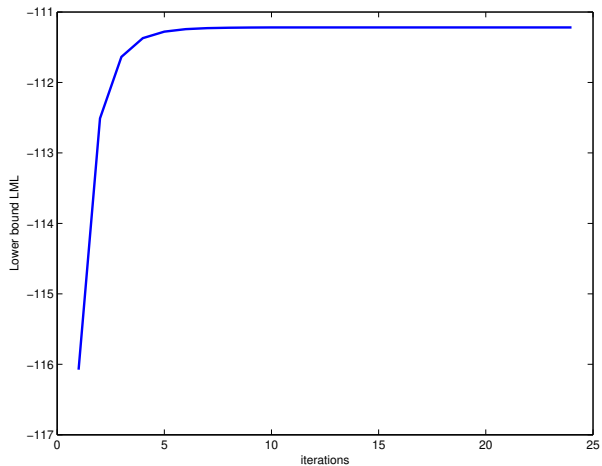
and

$$\tilde{\mu}_u = X \tilde{\mu}_\beta + \frac{\phi(X \tilde{\mu}_\beta)}{\Phi(X \tilde{\mu}_\beta)^y [\Phi(X \tilde{\mu}_\beta) - 1_n]^{1_n - y}}.$$

Probit example (n=200 observations)



Probit example



VI and exponential families

- **Exponential family** with sufficient statistics $\mathbf{t}(\mathbf{x})$

$$p(\mathbf{x}|\boldsymbol{\theta}) = h(\mathbf{x}) \exp \left\{ \boldsymbol{\eta}(\boldsymbol{\theta})^\top \mathbf{t}(\mathbf{x}) - a(\boldsymbol{\theta}) \right\}$$

- Suppose full conditional posterior is in the exponential family

$$p(\theta_j | \boldsymbol{\theta}_{-j}, \mathbf{y}) = h(\theta_j) \exp \{ \eta_j(\boldsymbol{\theta}_{-j}, \mathbf{y}) \theta_j - a(\eta_j(\boldsymbol{\theta}_{-j}, \mathbf{y})) \}$$

- Mean-field VI update

$$\begin{aligned} q(\theta_j) &\propto \exp \{ \mathbb{E}_{-j} \log p(\theta_j | \boldsymbol{\theta}_{-j}, \mathbf{y}) \} \\ &= \exp \{ \log h(\theta_j) + \mathbb{E}_{-j} [\eta_j(\boldsymbol{\theta}_{-j}, \mathbf{y})] \theta_j - \mathbb{E}_{-j} [a(\eta_j(\boldsymbol{\theta}_{-j}, \mathbf{y}))] \} \\ &\propto h(\theta_j) \exp \{ \mathbb{E}_{-j} [\eta_j(\boldsymbol{\theta}_{-j}, \mathbf{y})] \theta_j \} \end{aligned}$$

- Each $q(\theta_j)$ has same exponential family as its full conditional but with parameter $\mathbb{E}_{-j} [\eta_j(\boldsymbol{\theta}_{-j}, \mathbf{y})]$.

Digression - Conjugate prior for expon family

- **Exponential family** in the canonical parametrization

$$p(x|\theta) = h(x) \exp \left(\theta^\top t(x) - A(\theta) \right)$$

- **Likelihood**

$$p(x_1, \dots, x_n|\theta) = \left[\prod_{i=1}^n h(x_i) \right] \exp \left(\theta^\top \sum_{i=1}^n t(x_i) - nA(\theta) \right)$$

- **Conjugate prior**

$$p(\theta) = H(\tau_0, n_0) \exp \left(\theta^\top \tau_0 - n_0 A(\theta) \right),$$

where τ_0 and n_0 are prior hyperparameters and $H(\tau_0, n_0)$ is the normalizing constant which is known to exist if $n_0 > 0$.

Digression - Posterior in exponential family

■ Conjugate prior

$$p(\boldsymbol{\theta}) = H(\boldsymbol{\tau}_0, n_0) \exp \left(\boldsymbol{\theta}^\top \boldsymbol{\tau}_0 - n_0 A(\boldsymbol{\theta}) \right)$$

■ Posterior

$$p(\boldsymbol{\theta} | x_1, \dots, x_n) \propto \exp \left[\boldsymbol{\theta}^\top \left(\boldsymbol{\tau}_0 + \sum_{i=1}^n \mathbf{t}(x_i) \right) - (n_0 + n) A(\boldsymbol{\theta}) \right]$$

■ Prior-to-posterior updating

$$\boldsymbol{\tau}_0 \implies \boldsymbol{\tau}_n = \boldsymbol{\tau}_0 + \sum_{i=1}^n \mathbf{t}(x_i)$$

$$n_0 \implies n_0 + n$$

Digression - Bernoulli as exponential family

- **Exponential family** in the non-canonical parametrization

$$p(x|\theta) = h(x) \exp \left(\boldsymbol{\phi}(\theta)^\top \mathbf{t}(x) - A(\theta) \right)$$

- **Conjugate prior**

$$p(\theta) = H(\tau_0, n_0) \exp \left(\boldsymbol{\phi}(\theta)^\top \boldsymbol{\tau}_0 - n_0 A(\theta) \right)$$

- **Bernoulli likelihood**

$$\begin{aligned} p(x_1, \dots, x_n | \theta) &= \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i} \\ &= \exp \left(\log \left(\frac{\theta}{1-\theta} \right) \sum_{i=1}^n x_i - n \log \left(\frac{1}{1-\theta} \right) \right) \\ &= \exp \left(\boldsymbol{\phi}(\theta) \sum_{i=1}^n x_i - n A(\theta) \right) \end{aligned}$$

where $\boldsymbol{\phi} = \log \left(\frac{\theta}{1-\theta} \right)$ and $A(\theta) = \log \left(\frac{1}{1-\theta} \right)$.

- **Conjugate prior** $p(\boldsymbol{\phi})$

$$\exp(\boldsymbol{\phi}(\theta) \boldsymbol{\tau}_0 - n_0 A(\theta)) = \exp \left(\log \left(\frac{\theta}{1-\theta} \right) \tau_0 - n_0 \log \left(\frac{1}{1-\theta} \right) \right) = \theta^{\tau_0} (1 - \theta)^{n_0 - \tau_0}$$

which corresponds to an imaginary sample with τ_0 successes in n_0 trials.

Stochastic variational inference, Blei et al 2017

- **Mixture:** $\Pr(z_i = k) = \omega_k$ and $x_i | (z_i = k) \sim N(x | \mu_k, \sigma_k^2)$.

$$p(\mathbf{x}, \mathbf{z}, \boldsymbol{\mu}, \boldsymbol{\sigma}, \boldsymbol{\omega}) = p(\boldsymbol{\mu}, \boldsymbol{\sigma}, \boldsymbol{\omega}) \prod_{i=1}^n p(x_i | z_i, \boldsymbol{\mu}, \boldsymbol{\sigma}) p(z_i | \boldsymbol{\omega})$$

- **Global parameters:** $\boldsymbol{\theta} = (\boldsymbol{\mu}, \boldsymbol{\sigma}, \boldsymbol{\omega})^T$.
- **Local parameters:** z_i (**latents**). z_i is local to x_i .
- **Mean field VI** for **local parameter models** iterates:
 - ▶ Update the variational factor $q(\boldsymbol{\theta} | \boldsymbol{\lambda})$ for global parameters.
 - ▶ Update the variational factor $q(z_i | \varphi_i)$ for each local z_i .
- **Stochastic VI** (Blei et al 2017) for large data with latents:
 - ▶ Subsample a data point $s \in \{1, \dots, n\}$ and update $q(z_s | \varphi_s)$.
 - ▶ Update the variational factor $q(\boldsymbol{\theta} | \boldsymbol{\lambda})$ for global parameters.